

The University of the State of New York  
REGENTS HIGH SCHOOL EXAMINATION

# ALGEBRA I (Common Core)

Thursday, June 16, 2016 — 9:15 a.m. to 12:15 p.m., only

Student Name: \_\_\_\_\_

School Name: \_\_\_\_\_

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for Part I has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 37 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in Parts II, III, and IV directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

# High School Math Reference Sheet

- 1 inch = 2.54 centimeters
- 1 kilometer = 0.62 mile
- 1 cup = 8 fluid ounces
- 1 meter = 39.37 inches
- 1 pound = 16 ounces
- 1 pint = 2 cups
- 1 mile = 5280 feet
- 1 pound = 0.454 kilogram
- 1 quart = 2 pints
- 1 mile = 1760 yards
- 1 kilogram = 2.2 pounds
- 1 gallon = 4 quarts
- 1 mile = 1.609 kilometers
- 1 ton = 2000 pounds
- 1 gallon = 3.785 liters
- 1 liter = 0.264 gallon
- 1 liter = 1000 cubic centimeters

|                |                             |
|----------------|-----------------------------|
| Triangle       | $A = \frac{1}{2}bh$         |
| Parallelogram  | $A = bh$                    |
| Circle         | $A = \pi r^2$               |
| Circle         | $C = \pi d$ or $C = 2\pi r$ |
| General Prisms | $V = Bh$                    |
| Cylinder       | $V = \pi r^2 h$             |
| Sphere         | $V = \frac{4}{3}\pi r^3$    |
| Cone           | $V = \frac{1}{3}\pi r^2 h$  |
| Pyramid        | $V = \frac{1}{3}Bh$         |

|                          |                                                      |
|--------------------------|------------------------------------------------------|
| Pythagorean Theorem      | $a^2 + b^2 = c^2$                                    |
| Quadratic Formula        | $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$             |
| Arithmetic Sequence      | $a_n = a_1 + (n - 1)d$                               |
| Geometric Sequence       | $a_n = a_1 r^{n - 1}$                                |
| Geometric Series         | $S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$ |
| Radians                  | 1 radian = $\frac{180}{\pi}$ degrees                 |
| Degrees                  | 1 degree = $\frac{\pi}{180}$ radians                 |
| Exponential Growth/Decay | $A = A_0 e^{k(t - t_0)} + B_0$                       |

## Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for  
computations.

1 The expression  $x^4 - 16$  is equivalent to

- (1)  $(x^2 + 8)(x^2 - 8)$                       (3)  $(x^2 + 4)(x^2 - 4)$   
(2)  $(x^2 - 8)(x^2 - 8)$                       (4)  $(x^2 - 4)(x^2 - 4)$

2 An expression of the fifth degree is written with a leading coefficient of seven and a constant of six. Which expression is correctly written for these conditions?

- (1)  $6x^5 + x^4 + 7$                       (3)  $6x^7 - x^5 + 5$   
(2)  $7x^6 - 6x^4 + 5$                       (4)  $7x^5 + 2x^2 + 6$

3 The table below shows the year and the number of households in a building that had high-speed broadband internet access.

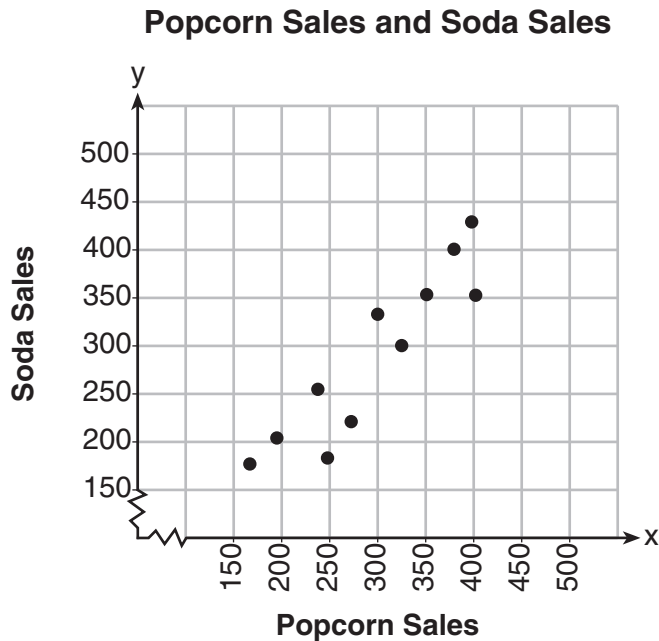
| Number of Households | 11   | 16   | 23   | 33   | 42   | 47   |
|----------------------|------|------|------|------|------|------|
| Year                 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 |

For which interval of time was the average rate of change the *smallest*?

- (1) 2002 – 2004                      (3) 2004 – 2006  
(2) 2003 – 2005                      (4) 2005 – 2007

Use this space for  
computations.

- 4 The scatterplot below compares the number of bags of popcorn and the number of sodas sold at each performance of the circus over one week.



Which conclusion can be drawn from the scatterplot?

- (1) There is a negative correlation between popcorn sales and soda sales.
  - (2) There is a positive correlation between popcorn sales and soda sales.
  - (3) There is no correlation between popcorn sales and soda sales.
  - (4) Buying popcorn causes people to buy soda.
- 5 The Celluloid Cinema sold 150 tickets to a movie. Some of these were child tickets and the rest were adult tickets. A child ticket cost \$7.75 and an adult ticket cost \$10.25. If the cinema sold \$1470 worth of tickets, which system of equations could be used to determine how many adult tickets,  $a$ , and how many child tickets,  $c$ , were sold?

- |                         |                         |
|-------------------------|-------------------------|
| (1) $a + c = 150$       | (3) $a + c = 150$       |
| $10.25a + 7.75c = 1470$ | $7.75a + 10.25c = 1470$ |
| (2) $a + c = 1470$      | (4) $a + c = 1470$      |
| $10.25a + 7.75c = 150$  | $7.75a + 10.25c = 150$  |

**Use this space for  
computations.**

- 6** The tables below show the values of four different functions for given values of  $x$ .

| x | f(x) |
|---|------|
| 1 | 12   |
| 2 | 19   |
| 3 | 26   |
| 4 | 33   |

| $x$ | $g(x)$ |
|-----|--------|
| 1   | -1     |
| 2   | 1      |
| 3   | 5      |
| 4   | 13     |

| x | h(x) |
|---|------|
| 1 | 9    |
| 2 | 12   |
| 3 | 17   |
| 4 | 24   |

| x | k(x) |
|---|------|
| 1 | -2   |
| 2 | 4    |
| 3 | 14   |
| 4 | 28   |

Which table represents a linear function?

- |            |            |
|------------|------------|
| (1) $f(x)$ | (3) $h(x)$ |
| (2) $g(x)$ | (4) $k(x)$ |

- 7** The acidity in a swimming pool is considered normal if the average of three pH readings,  $p$ , is defined such that  $7.0 < p < 7.8$ . If the first two readings are 7.2 and 7.6, which value for the third reading will result in an overall rating of normal?

- |         |         |
|---------|---------|
| (1) 6.2 | (3) 8.6 |
| (2) 7.3 | (4) 8.8 |

- 8** Dan took 12.5 seconds to run the 100-meter dash. He calculated the time to be approximately

- (1) 0.2083 minute                      (3) 0.2083 hour  
(2) 750 minutes                        (4) 0.52083 hour

- 9** When  $3x + 2 \leq 5(x - 4)$  is solved for  $x$ , the solution is

- $$\begin{array}{ll} (1) & x \leq 3 \\ (2) & x \geq 3 \\ (3) & x \leq -11 \\ (4) & x \geq 11 \end{array}$$

**10** The expression  $3(x^2 - 1) - (x^2 - 7x + 10)$  is equivalent to

- (1)  $2x^2 - 7x + 7$                       (3)  $2x^2 - 7x + 9$   
(2)  $2x^2 + 7x - 13$                       (4)  $2x^2 + 7x - 11$

**11** The range of the function  $f(x) = x^2 + 2x - 8$  is all real numbers

- (1) less than or equal to  $-9$   
(2) greater than or equal to  $-9$   
(3) less than or equal to  $-1$   
(4) greater than or equal to  $-1$

**12** The zeros of the function  $f(x) = x^2 - 5x - 6$  are

- (1)  $-1$  and  $6$                               (3)  $2$  and  $-3$   
(2)  $1$  and  $-6$                               (4)  $-2$  and  $3$

**13** In a sequence, the first term is  $4$  and the common difference is  $3$ .  
The fifth term of this sequence is

- (1)  $-11$                                       (3)  $16$   
(2)  $-8$                                         (4)  $19$

**14** The growth of a certain organism can be modeled by  $C(t) = 10(1.029)^{24t}$ , where  $C(t)$  is the total number of cells after  $t$  hours. Which function is approximately equivalent to  $C(t)$ ?

- (1)  $C(t) = 240(.083)^{24t}$                       (3)  $C(t) = 10(1.986)^t$   
(2)  $C(t) = 10(.083)^t$                               (4)  $C(t) = 240(1.986)^{\frac{t}{24}}$

**Use this space for  
computations.**

- 15** A public opinion poll was taken to explore the relationship between age and support for a candidate in an election. The results of the poll are summarized in the table below.

| Age     | For | Against | No Opinion |
|---------|-----|---------|------------|
| 21–40   | 30  | 12      | 8          |
| 41–60   | 20  | 40      | 15         |
| Over 60 | 25  | 35      | 15         |

What percent of the 21–40 age group was for the candidate?

- |        |        |
|--------|--------|
| (1) 15 | (3) 40 |
| (2) 25 | (4) 60 |

- 16** Which equation and ordered pair represent the correct vertex form and vertex for  $j(x) = x^2 - 12x + 7$ ?

- $$\begin{aligned} (1) \quad & j(x) = (x - 6)^2 + 43, & (6, 43) \\ (2) \quad & j(x) = (x - 6)^2 + 43, & (-6, 43) \\ (3) \quad & j(x) = (x - 6)^2 - 29, & (6, -29) \\ (4) \quad & j(x) = (x - 6)^2 - 29, & (-6, -29) \end{aligned}$$

- 17** A student invests \$500 for 3 years in a savings account that earns 4% interest per year. No further deposits or withdrawals are made during this time. Which statement does *not* yield the correct balance in the account at the end of 3 years?

- (1)  $500(1.04)^3$
- (2)  $500(1 - .04)^3$
- (3)  $500(1 + .04)(1 + .04)(1 + .04)$
- (4)  $500 + 500(.04) + 520(.04) + 540.8(.04)$

Use this space for  
computations.

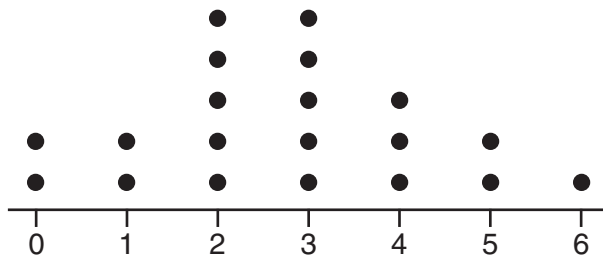
- 18 The line represented by the equation  $4y + 2x = 33.6$  shares a solution point with the line represented by the table below.

| x  | y   |
|----|-----|
| -5 | 3.2 |
| -2 | 3.8 |
| 2  | 4.6 |
| 4  | 5   |
| 11 | 6.4 |

The solution for this system is

- (1)  $(-14.0, -1.4)$                       (3)  $(1.9, 4.6)$   
(2)  $(-6.8, 5.0)$                         (4)  $(6.0, 5.4)$
- 19 What is the solution of the equation  $2(x + 2)^2 - 4 = 28$ ?
- (1) 6, only                                      (3) 2 and -6  
(2) 2, only                                      (4) 6 and -2

- 20 The dot plot shown below represents the number of pets owned by students in a class.



Which statement about the data is *not* true?

- (1) The median is 3.  
(2) The interquartile range is 2.  
(3) The mean is 3.  
(4) The data contain no outliers.



**21** What is the largest integer,  $x$ , for which the value of  $f(x) = 5x^4 + 30x^2 + 9$  will be greater than the value of  $g(x) = 3^x$ ?

- (1) 7
- (2) 8
- (3) 9
- (4) 10

**22** The graphs of the functions  $f(x) = |x - 3| + 1$  and  $g(x) = 2x + 1$  are drawn. Which statement about these functions is true?

- (1) The solution to  $f(x) = g(x)$  is 3.
- (2) The solution to  $f(x) = g(x)$  is 1.
- (3) The graphs intersect when  $y = 1$ .
- (4) The graphs intersect when  $x = 3$ .

**23** A store sells self-serve frozen yogurt sundaes. The function  $C(w)$  represents the cost, in dollars, of a sundae weighing  $w$  ounces. An appropriate domain for the function would be

- (1) integers
- (2) rational numbers
- (3) nonnegative integers
- (4) nonnegative rational numbers

**24** Sara was asked to solve this word problem: “The product of two consecutive integers is 156. What are the integers?”

What type of equation should she create to solve this problem?

- (1) linear
  - (2) quadratic
  - (3) exponential
  - (4) absolute value
-

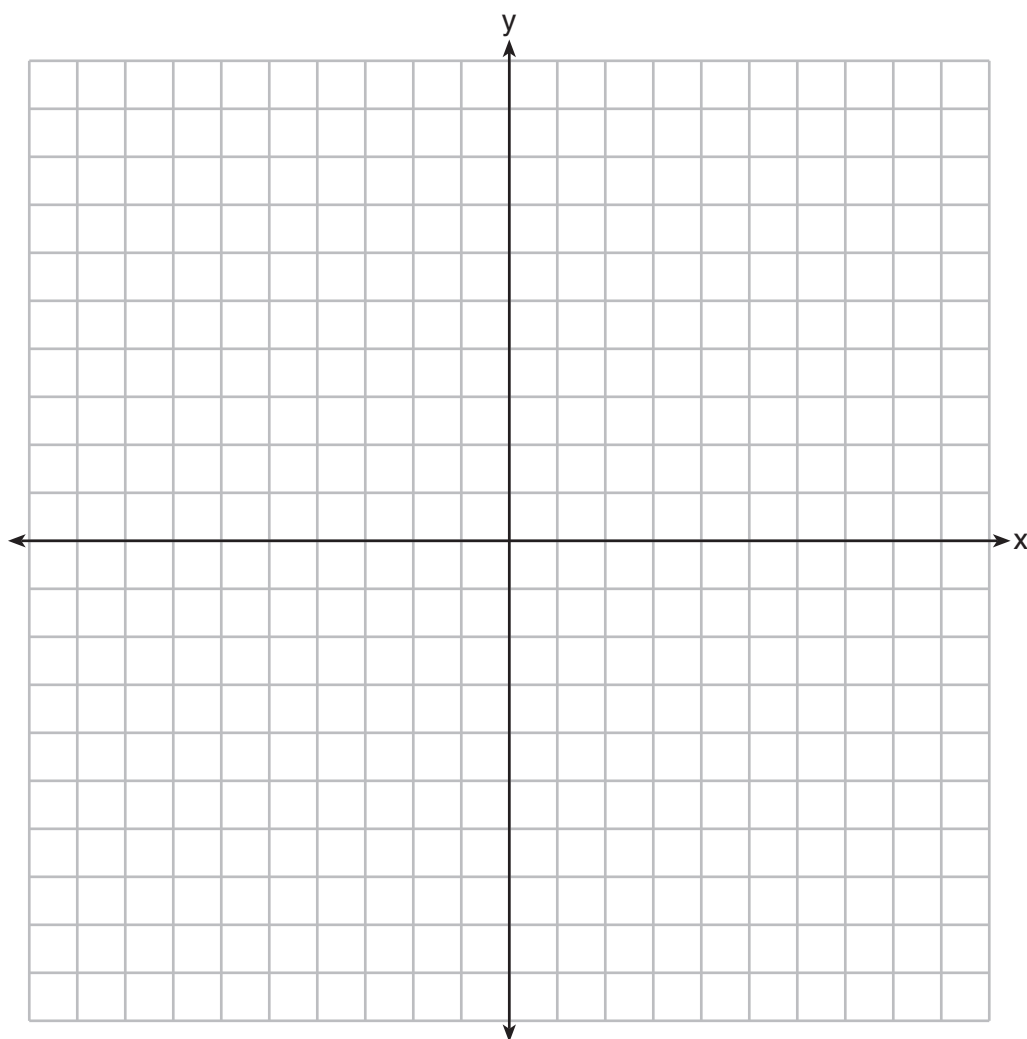
## Part II

Answer all 8 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

**25** Given that  $f(x) = 2x + 1$ , find  $g(x)$  if  $g(x) = 2[f(x)]^2 - 1$ .

**26** Determine if the product of  $3\sqrt{2}$  and  $8\sqrt{18}$  is rational or irrational. Explain your answer.

**27** On the set of axes below, draw the graph of  $y = x^2 - 4x - 1$ .



State the equation of the axis of symmetry.

**28** Amy solved the equation  $2x^2 + 5x - 42 = 0$ . She stated that the solutions to the equation were  $\frac{7}{2}$  and  $-6$ . Do you agree with Amy's solutions? Explain why or why not.

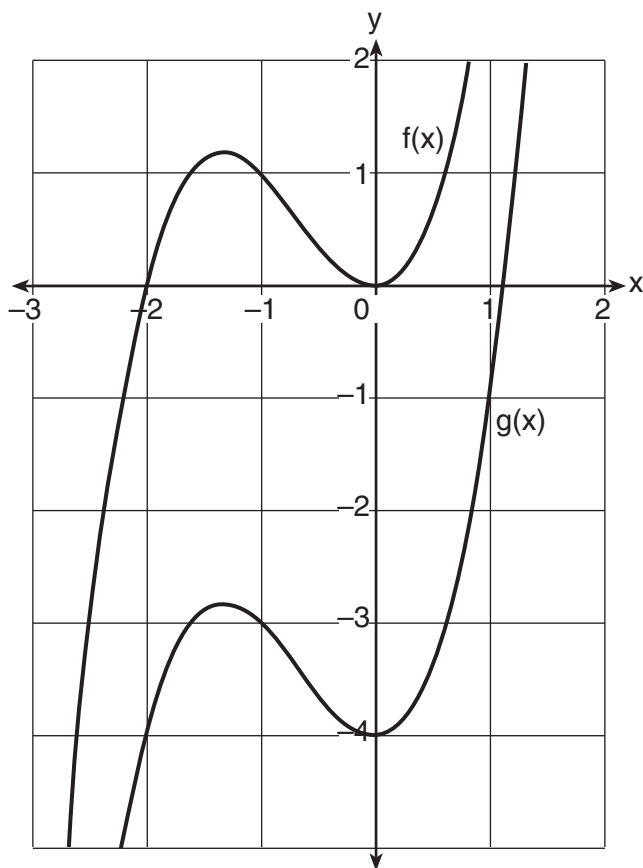
**29** Sue and Kathy were doing their algebra homework. They were asked to write the equation of the line that passes through the points  $(-3,4)$  and  $(6,1)$ . Sue wrote  $y - 4 = -\frac{1}{3}(x + 3)$  and Kathy wrote  $y = -\frac{1}{3}x + 3$ . Justify why both students are correct.

**30** During a recent snowstorm in Red Hook, NY, Jaime noted that there were 4 inches of snow on the ground at 3:00 p.m., and there were 6 inches of snow on the ground at 7:00 p.m.

If she were to graph these data, what does the slope of the line connecting these two points represent in the context of this problem?

- 31** The formula for the sum of the degree measures of the interior angles of a polygon is  $S = 180(n - 2)$ . Solve for  $n$ , the number of sides of the polygon, in terms of  $S$ .

- 32 In the diagram below,  $f(x) = x^3 + 2x^2$  is graphed. Also graphed is  $g(x)$ , the result of a translation of  $f(x)$ .



Determine an equation of  $g(x)$ . Explain your reasoning.



### Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

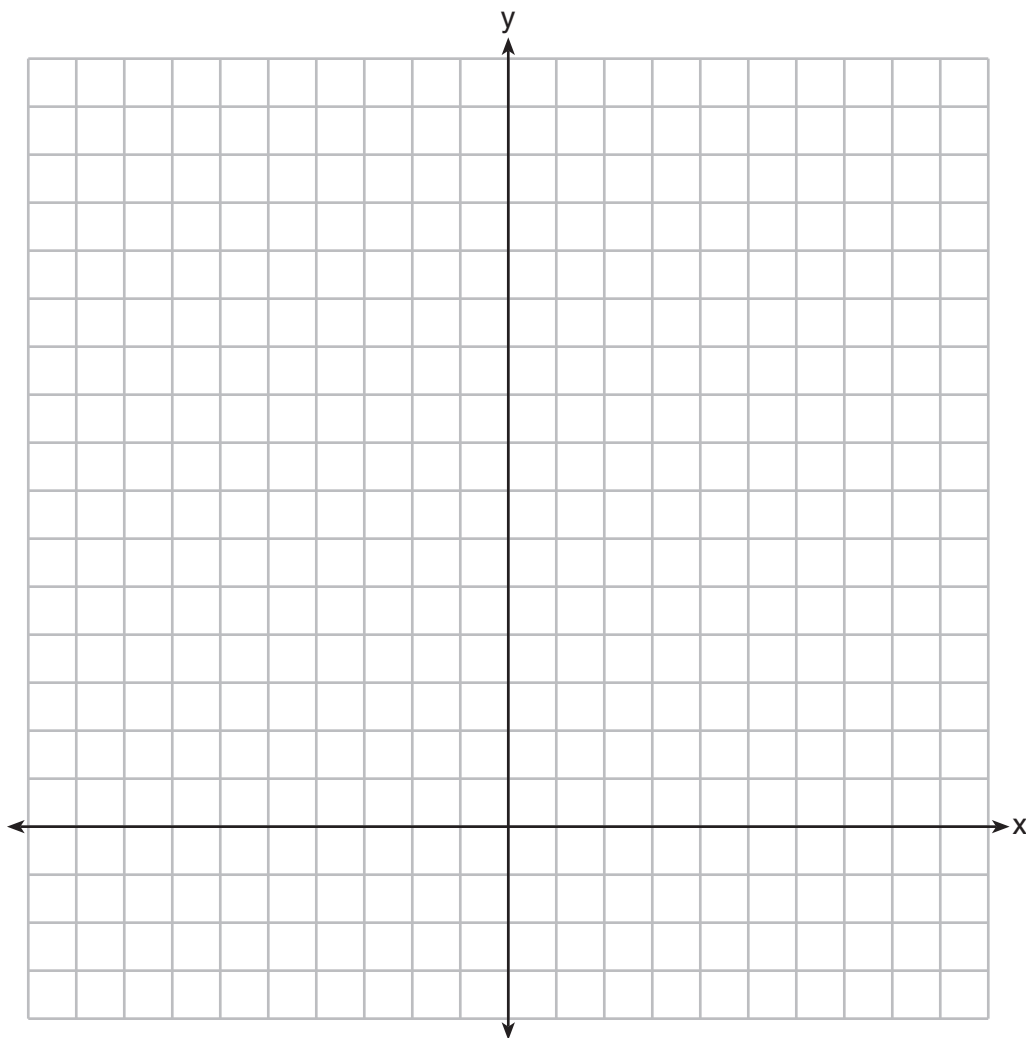
- 33 The height,  $H$ , in feet, of an object dropped from the top of a building after  $t$  seconds is given by  $H(t) = -16t^2 + 144$ .

How many feet did the object fall between one and two seconds after it was dropped?

Determine, algebraically, how many seconds it will take for the object to reach the ground.

- 34 The sum of two numbers,  $x$  and  $y$ , is more than 8. When you double  $x$  and add it to  $y$ , the sum is less than 14.

Graph the inequalities that represent this scenario on the set of axes below.



Kai says that the point  $(6,2)$  is a solution to this system. Determine if he is correct and explain your reasoning.

**35** An airplane leaves New York City and heads toward Los Angeles. As it climbs, the plane gradually increases its speed until it reaches cruising altitude, at which time it maintains a constant speed for several hours as long as it stays at cruising altitude. After flying for 32 minutes, the plane reaches cruising altitude and has flown 192 miles. After flying for a total of 92 minutes, the plane has flown a total of 762 miles.

Determine the speed of the plane, at cruising altitude, in miles per minute.

Write an equation to represent the number of miles the plane has flown,  $y$ , during  $x$  minutes at cruising altitude, only.

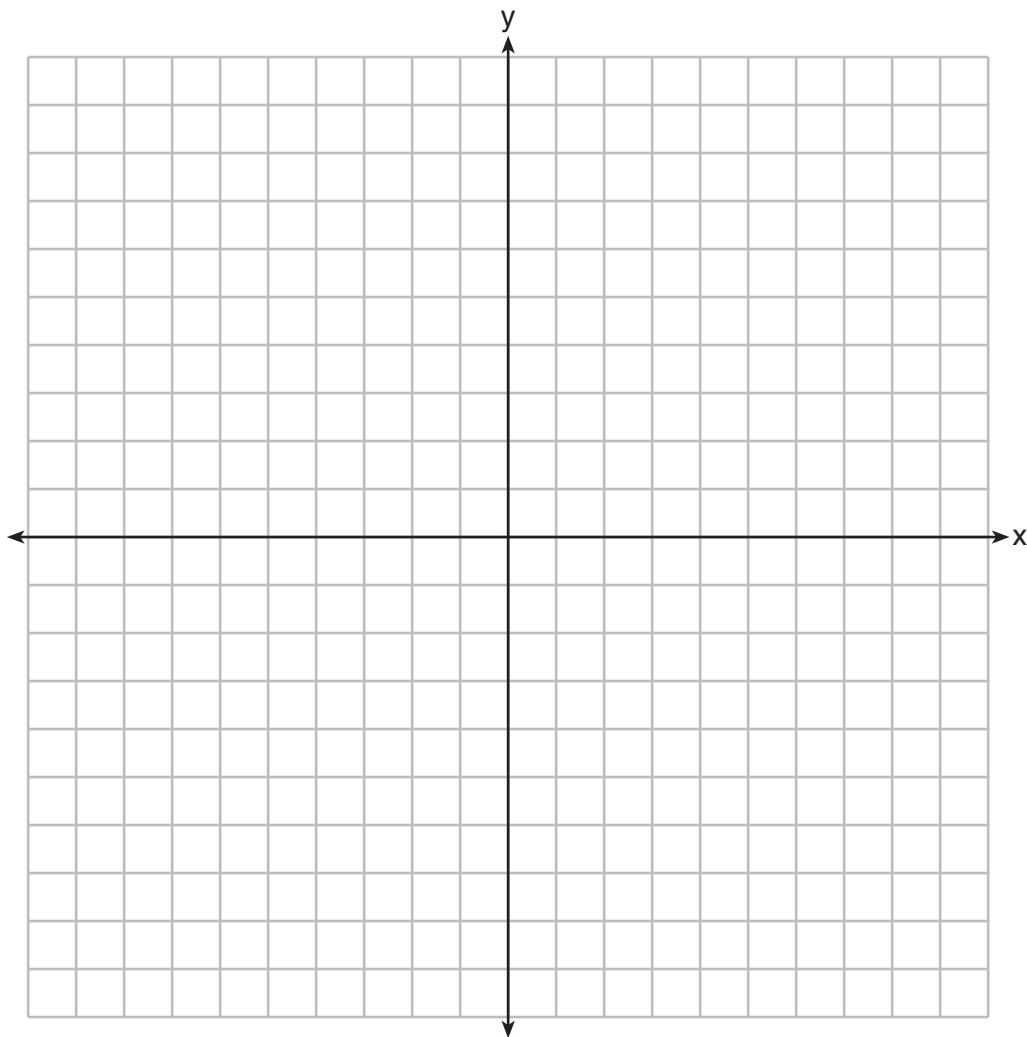
Assuming that the plane maintains its speed at cruising altitude, determine the total number of miles the plane has flown 2 hours into the flight.

**36** On the set of axes below, graph

$$g(x) = \frac{1}{2}x + 1$$

and

$$f(x) = \begin{cases} 2x + 1, & x \leq -1 \\ 2 - x^2, & x > -1 \end{cases}$$



How many values of  $x$  satisfy the equation  $f(x) = g(x)$ ? Explain your answer, using evidence from your graphs.

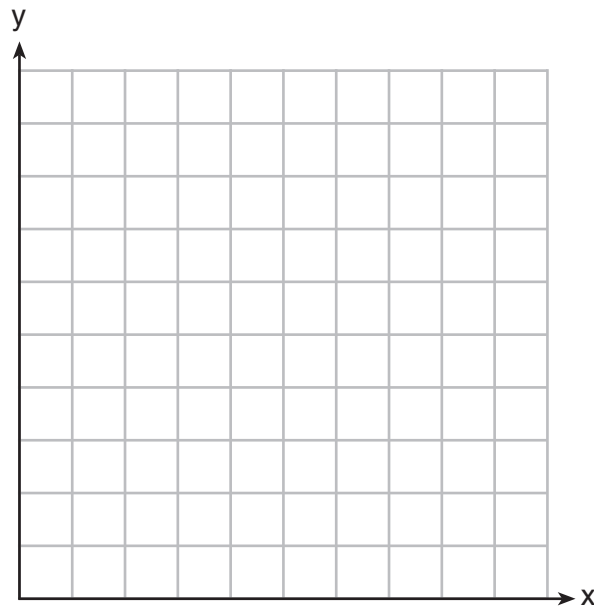
#### Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

- 37 Franco and Caryl went to a bakery to buy desserts. Franco bought 3 packages of cupcakes and 2 packages of brownies for \$19. Caryl bought 2 packages of cupcakes and 4 packages of brownies for \$24. Let  $x$  equal the price of one package of cupcakes and  $y$  equal the price of one package of brownies.

Write a system of equations that describes the given situation.

On the set of axes below, graph the system of equations.



Determine the exact cost of one package of cupcakes and the exact cost of one package of brownies in dollars and cents. Justify your solution.

If the student's responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

## Part I

Allow a total of 48 credits, 2 credits for each of the following.

|                           |                            |                            |
|---------------------------|----------------------------|----------------------------|
| (1) . . . . . 3 . . . . . | (9) . . . . . 4 . . . . .  | (17) . . . . . 2 . . . . . |
| (2) . . . . . 4 . . . . . | (10) . . . . . 2 . . . . . | (18) . . . . . 4 . . . . . |
| (3) . . . . . 1 . . . . . | (11) . . . . . 2 . . . . . | (19) . . . . . 3 . . . . . |
| (4) . . . . . 2 . . . . . | (12) . . . . . 1 . . . . . | (20) . . . . . 3 . . . . . |
| (5) . . . . . 1 . . . . . | (13) . . . . . 3 . . . . . | (21) . . . . . 3 . . . . . |
| (6) . . . . . 1 . . . . . | (14) . . . . . 3 . . . . . | (22) . . . . . 2 . . . . . |
| (7) . . . . . 2 . . . . . | (15) . . . . . 4 . . . . . | (23) . . . . . 4 . . . . . |
| (8) . . . . . 1 . . . . . | (16) . . . . . 3 . . . . . | (24) . . . . . 2 . . . . . |

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the "Model Response Set," for the Regents Examination in Algebra I (Common Core). This guidance is recommended to be part of the scorer training. Schools are encouraged to incorporate the Model Response Sets into the scorer training or to use them as additional information during scoring. While not reflective of all scenarios, the model responses selected for the Model Response Set illustrate how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department's web site at <http://www.nysedregents.org/algebraone/>.

## Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2]  $2(2x + 1)^2 - 1$  or an equivalent expression is written.

[1] Appropriate work is shown, but one computational error is made.

**or**

[1] Appropriate work is shown, but one conceptual error is made.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] Rational, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

**or**

[1] Appropriate work is shown, but one conceptual error is made.

**or**

[1] A correct explanation is written, but rational is not stated.

**or**

[1] 144 and rational, but the explanation is missing or incorrect.

[0] Rational, but the explanation is missing or incorrect.

**or**

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (27) [2] A correct graph is drawn and  $x = 2$  is stated.
- [1] Appropriate work is shown, but one computational or graphing error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] A correct graph is drawn, but no further correct work is shown.
- or**
- [1]  $x = 2$  is stated, but no further correct work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (28) [2] Yes, and a correct explanation is written.
- [1] Appropriate work is shown, but one computational or factoring error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] Yes, but the explanation is incomplete.
- [0] Yes, but the explanation is missing or incorrect.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (29) [2] A correct justification is given.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] An incomplete justification is given.
- [0] An incorrect justification is given.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.



- (30) [2] The number of inches of snow falling per hour or an equivalent explanation is written.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] An incomplete explanation is written.
- or**
- [1] Appropriate work is shown to find  $\frac{1}{2}$ , but no explanation or an incorrect explanation is written.
- or**
- [1] An explanation that doesn't refer to the context is written.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- 
- (31) [2]  $n = \frac{S+360}{180}$  or  $n = \frac{S}{180} + 2$ , and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1]  $n = \frac{S+360}{180}$ , but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(32) [2]  $g(x) = x^3 + 2x^2 - 4$  or  $g(x) = f(x) - 4$ , and a correct explanation is written.

[1] Appropriate work is shown, but one conceptual error is made.

***or***

[1] Appropriate work is shown, but the explanation is missing or incorrect.

***or***

[1] A correct explanation is written, but no further correct work is shown.

***or***

[1] The expression  $x^3 + 2x^2 - 4$  is written, and a correct explanation is written.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

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### Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(33) [4] 48, 3, and correct work is shown.

[3] Appropriate work is shown, but one computational error is made.

[2] Appropriate work is shown, but two or more computational errors are made.

***or***

[2] Appropriate work is shown, but one conceptual error is made.

***or***

[2] Appropriate work is shown to find 48 or 3, but no further correct work is shown.

[1] Appropriate work is shown, but one conceptual error and one computational error are made.

***or***

[1] Appropriate work is shown to find 128 and 80, but no further correct work is shown.

***or***

[1] 48 and 3, but no work is shown.

[0] 128 and 80, but no work is shown.

***or***

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] Both inequalities are graphed correctly and at least one is labeled, and a correct explanation stating he is incorrect is written.
- [3] Appropriate work is shown, but one computational, graphing, or labeling error is made.
- or***
- [3] Appropriate work is shown, but the explanation is missing or incorrect.
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown, but two or more computational, graphing, or labeling errors are made.
- or***
- [2] Both inequalities are graphed correctly, but no further correct work is shown.
- [1] Both inequalities are stated correctly, but no further correct work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (35) [4] 9.5,  $y = 9.5x$  and 1028, and correct work is shown.
- [3] Appropriate work is shown, but one computational error is made.
- or***
- [3] Appropriate work is shown to find 9.5 and 1028, but no equation or an incorrect equation is written.
- or***
- [3] Appropriate work is shown to find 9.5,  $y = 9.5x$  and 266, the number of miles flown after 92 minutes, but no further correct work is shown.
- [2] Appropriate work is shown, but two or more computational errors are made.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown to find 9.5 and  $y = 9.5x$ , but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational error are made.
- or***
- [1] Appropriate work is shown to find 9.5, but no further correct work is shown.
- or***
- [1] 9.5,  $y = 9.5x$  and 1028, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [4] Correct graphs are drawn, 1, and a correct explanation is written.
- [3] Appropriate work is shown, but one computational or graphing error is made.
- or**
- [3] Appropriate work is shown, but the explanation is missing or incorrect.
- [2] Appropriate work is shown, but two or more computational or graphing errors are made.
- or**
- [2] Appropriate work is shown, but one conceptual error is made.
- or**
- [2] Correct graphs are drawn, but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational or graphing error are made.
- or**
- [1] Two of the following equations are graphed correctly:  
 $y = 2x + 1$  over  $x \leq -1$ ,  $y = 2 - x^2$  over  $x > -1$ , or  $y = \frac{1}{2}x + 1$  for all real numbers.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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## Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(37) [6]  $3x + 2y = 19$  and  $2x + 4y = 24$  are written and graphed correctly, and at least one is labeled,  $x = 3.50$  and  $y = 4.25$  or the coordinates  $(3.50, 4.25)$  are stated, and a justification is given.

[5] Appropriate work is shown, but one computational, graphing, or labeling error is made.

**or**

[5] Appropriate work is shown, but the justification is missing.

**or**

[5] Appropriate work is shown, but only one cost is stated correctly.

[4] Appropriate work is shown, but two computational, graphing, or labeling errors are made.

**or**

[4] Appropriate work is shown, but one conceptual error is made.

**or**

[4] A correct system of equations is written and graphed correctly, and at least one is labeled, but no further correct work is shown.

**or**

[4] Appropriate work is shown, but no graphs are drawn.

[3] Appropriate work is shown, but three or more computational, graphing, or labeling errors are made.

**or**

[3] Appropriate work is shown, but one conceptual error and one computational, graphing, or labeling error are made.

**or**

[3] A correct system of equations is written, and one line is graphed and labeled correctly, but no further correct work is shown.

[2] Appropriate work is shown, but one conceptual error and two or more computational, graphing, or labeling errors are made.

**or**

[2] Appropriate work is shown, but two conceptual errors are made.

***or***

[2] A correct system of equations is written, but no further correct work is shown.

***or***

[2] Only one equation is written and graphed correctly, but no further correct work is shown.

[1] Appropriate work is shown, but two conceptual errors and one computational, graphing, or labeling error are made.

***or***

[1] Only one of the equations is written correctly, but no further correct work is shown.

***or***

[1]  $x = 3.50$  and  $y = 4.25$  or the coordinates  $(3.50, 4.25)$  are stated, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

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**Map to the Common Core Learning Standards  
Algebra I (Common Core)  
June 2016**

| <b>Question</b> | <b>Type</b>     | <b>Credits</b> | <b>Cluster</b> |
|-----------------|-----------------|----------------|----------------|
| 1               | Multiple Choice | 2              | A-SSE.A        |
| 2               | Multiple Choice | 2              | A-SSE.A        |
| 3               | Multiple Choice | 2              | F-IF.B         |
| 4               | Multiple Choice | 2              | S-ID.C         |
| 5               | Multiple Choice | 2              | A-REI.C        |
| 6               | Multiple Choice | 2              | F-LE.A         |
| 7               | Multiple Choice | 2              | A-CED.A        |
| 8               | Multiple Choice | 2              | N-Q.A          |
| 9               | Multiple Choice | 2              | A-REI.B        |
| 10              | Multiple Choice | 2              | A-APR.A        |
| 11              | Multiple Choice | 2              | F-IF.A         |
| 12              | Multiple Choice | 2              | A-APR.B        |
| 13              | Multiple Choice | 2              | F-IF.A         |
| 14              | Multiple Choice | 2              | A-SSE.B        |
| 15              | Multiple Choice | 2              | S-ID.B         |
| 16              | Multiple Choice | 2              | F-IF.C         |
| 17              | Multiple Choice | 2              | F-BF.A         |
| 18              | Multiple Choice | 2              | F-IF.C         |
| 19              | Multiple Choice | 2              | A-REI.B        |
| 20              | Multiple Choice | 2              | S-ID.A         |

|    |                      |   |         |
|----|----------------------|---|---------|
| 21 | Multiple Choice      | 2 | F-LE.A  |
| 22 | Multiple Choice      | 2 | F-IF.A  |
| 23 | Multiple Choice      | 2 | F-IF.B  |
| 24 | Multiple Choice      | 2 | A-CED.A |
| 25 | Constructed Response | 2 | F-IF.A  |
| 26 | Constructed Response | 2 | N-RN.B  |
| 27 | Constructed Response | 2 | A-REI.D |
| 28 | Constructed Response | 2 | A-REI.A |
| 29 | Constructed Response | 2 | A-REI.D |
| 30 | Constructed Response | 2 | S-ID.C  |
| 31 | Constructed Response | 2 | A-CED.A |
| 32 | Constructed Response | 2 | F-BF.B  |
| 33 | Constructed Response | 4 | A-SSE.B |
| 34 | Constructed Response | 4 | A-REI.D |
| 35 | Constructed Response | 4 | A-CED.A |
| 36 | Constructed Response | 4 | F-IF.C  |
| 37 | Constructed Response | 6 | A-CED.A |